



Live membrane filters with immobilized quorum quenching bacterial strains for anti-biofouling[☆]

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ABSTRACT

Membrane filters are core materials for water treatment, reuse, and desalination. However, their fouling caused by biofilm growth is a major challenge, needing resolution. Here, we report the new strategies and efficacy of anti-biofouling membranes fabricated through the immobilization of quorum quenching (QQ) bacterial strains (*Rhodococcus* sp. BH4). The QQ cells are not attached to the membrane by direct phase inversion. However, they anchor to the membrane with its incubation in the presence of hydrophilic polymers (polyvinyl alcohol and alginate) as evidenced in the spectroscopic and morphological observations. Membrane filters with live QQ bacteria can degrade a signal molecule (N-octanoyl-L-homoserine lactone) with the first-order rate constant of $0.58\text{--}0.82\text{ h}^{-1}$ and inhibit the growth of a biofilm-forming bacterium (PAO1) and its secretion of biopolymers. The QQ membrane demonstrates its significant anti-biofouling effect in bioreactors for treating synthetic wastewater (i.e., membrane fouling was delayed by 57–67% compared to the naked membrane), although the initial water permeabilities are reduced with surface modification by 34–47% compared to pristine condition. The findings of this work bring the broad potential for material fabrication requiring biofouling control, encountered in our daily lives and mechanical, marine, and medical industries in addition to membrane filters.

1. Introduction

The United Nations World Water Development Report states that as of 2018, 3.6 billion people suffer from water scarcity for at least a month in a year [1]. The World Health Organization also reported that in 2017, 29% of the world population lacked access to safely managed drinking water services [2]. Besides, anthropogenic pollution by effluents from domestic, agricultural, and industrial practices deteriorates water sources' quality [3,4]. These threats draw our attention to more practical and reliable treatment options. Membrane filters, which can separate and eliminate contaminants by rated pore sizes [5,6], are vital in water management, accounting for ~53% of the total number of world treatment processes [7]. Furthermore, membrane technologies are sustainable, environmentally friendly, flexible, and resilient in the circular economy because they can selectively extract valuable resources such as nutrients, metals, and energy from wastewater effluents in addition to water [3].

Precisely, a membrane bioreactor (MBR) technology, based on a synergistic combination of biodegradation and physical separation, is considered as one of the most efficient and popular approaches for wastewater reclamation and reuse, targeting alternative water resources [8,9]. Despite the advantages of the MBR technology, membrane fouling, particularly biofilm growth on the membrane filter (biofouling caused by quorum sensing (QS) mechanisms), is one of the most serious issues, needing resolution [8]. Several physicochemical methods have been developed to solve biofouling problems, such as air scouring [10], permeate backwashing [11], chemically enhanced backwashing [12], subcritical flux membrane filtration [13,14], membrane vibration [15], membrane reciprocation [16], or their combination [15]. However, biofouling, which naturally occurs on the solid's surface (membrane surface) via cell-to-cell communications (quorum sensing) [17,18], is inevitable, thus requiring novel approaches for anti-biofouling.

Since 2009, quenching bacterial signaling at the membrane surface has been reported using signal molecule-degrading enzymes, quorum

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sensing inhibiting chemicals, and quorum quenching (QQ) bacterial strains [19–22]. QQ disrupts microbial communications through the degradation of signal molecules or downregulation of their inducers or receptors and thereby prevents microbial biofilm formation. QQ bacterial strains (e.g., *Rhodococcus* sp. BH4) are effective in mitigating membrane biofouling when injected into a bioreactor in media form, regardless of either being suspended or fixed in the reactor [23]. Several types of media, such as vessel [24], bead [25], sheet [9,26], cylinder [27], hollow cylinder [28], rotating sheet [29], and core-shell structured bead [30], have been developed to enhance the QQ efficacy (anti-biofouling efficiency) when placed inside the bioreactor. However, the addition of exogenous media into the reactor brings about secondary issues, such as the entrapment of the media inside the membrane fibers and the media loss during reactor operation [23,26]. Thus, continual strong demand exists for the fabrication of a biofouling-resistant membrane devoid of media.

Immobilization of QQ enzymes (acylase or lactonase) on the membrane surface has been previously reported [31,32]. However, the enzymes' stability during their long-term use in a bioreactor is questionable because they are highly prone to biodegradation [33]. Direct immobilization of live QQ bacterial cells on a solid substrate (e.g., membrane surface) appears to be greatly interesting for anti-biofouling but has been unexplored because bacterial cells cannot survive in an organic solvent during membrane fabrication by phase inversion.

Therefore, in this study, we develop a novel method to immobilize the QQ bacteria (*Rhodococcus* sp. BH4) on polymeric membranes made from polysulfone (PS) and polyvinylidene difluoride (PVDF). Attempts to attach live BH4 strains to membrane filters are made by immersing the membranes in an aqueous solution containing BH4 cells and hydrophilic polymers. The latter functions as a binder to make the bacterial cells anchor to the membrane surface. The spectroscopic, morphological, and biological characteristics of membrane filters before and after the immobilization of QQ bacterial cells are evaluated. Finally, the anti-biofouling efficacy of live QQ cell-immobilized membranes is demonstrated in the bioreactor treatment of synthetic wastewater.

2. Materials and methods

2.1. Polymers and chemicals

PS pellets (Udel® P-3500 LCD) were purchased from Solvay Korea. Polyvinylpyrrolidone (PVP; average molecular weights = 58000) and N, N-dimethylacetamide anhydrous (99.8%) were purchased from Alfa Aesar Korea. Polyvinyl alcohol (PVA) and sodium alginate powder were purchased from Wako Pure Chemical (degree of polymerization = 2000) and Junsei Chemical, respectively. All other chemicals used were of guaranteed grade and used as received.

2.2. Hollow fiber membrane filters

For the synthesis of the PS hollow fiber membrane filters, a dope solution was prepared by dissolving 15-wt% PS pellets and 20-wt% PVP in 65-wt% dimethylacetamide (DMA) at 60 °C for 6 h with continuous stirring at 60 rpm [34]. The dope solution was then used to prepare the base hollow fiber membrane filters. More specifically, the dope solution was pumped through the outer side of a 1-mm inner diameter double orifice spinneret into a coagulation solution, containing a mixture of boric acid and calcium chloride (7:4 in mass ratio). A bore solution (70% v/v DMA in water) was ejected through the inside of the orifice spinneret at an extrusion rate of 1 mL/min using a syringe pump. Hollow fiber membranes were formed and soaked for 2 h in the coagulation solution and afterward, were stabilized in a 0.5-M sodium sulfate solution for an additional 8 h. In contrast, commercially available PVDF hollow fiber membrane filters (Cleanfil® S series) with a pore diameter of 0.1 µm were obtained from Kolon Industries (Korea).

2.3. Hydrophilic polymer coating and QQ bacterial strain immobilization

A coating solution containing 2-wt% PVA, 0.2-wt% sodium alginate, and 97.3-wt% deionized water was first prepared and then autoclaved for 15 min at 121 °C for sterilization. The coating solution was used for the preparation of vacant PS and PVDF membrane filters, which only have hydrophilic polymers on the top of the base membrane filters. Conversely, a QQ bacterium (0.5-wt% *Rhodococcus* sp. BH4) was added to the sterilized coating solution after cooling to room temperature for 30 min. The resulting mixture (hydrophilic polymers + BH4 strains, which is called a QQ solution) was used for the immobilization of QQ bacteria onto the base membrane (QQ membrane).

More specifically, three strategies to prepare QQ membranes were created as described in Fig. 1. For Strategy 1, the QQ solution containing only BH4 strains, with no hydrophilic polymers, was used and ejected through the most outer side of a triple orifice spinneret. For Strategy 2, the QQ solution contains both BH4 strains and hydrophilic polymers as described above. For the preparation of vacant membranes by Strategy 3 (Fig. 1c), the PS and PVDF membranes were immersed (dip-coated) in the hydrophilic polymer solution for 6 h. The coated membranes were then kept in the coagulation solution for 2 h and further stabilized in 0.5-M Na₂SO₄ solution for another 8 h. For the preparation of QQ membranes by Strategy 3, the PS and PVDF membranes were placed in a mixture of hydrophilic polymers and BH4 strains for 6 h. The membranes were then kept in the same manner in the coagulation solution and Na₂SO₄ solution as described above. For the membranes' preservation, the vacant and QQ membranes were immersed in a 25-wt% glycerol aqueous solution for 12 h. The membranes were then stored in the refrigerator at 4 °C before use. The QQ membrane was placed in a 2.5%-Luria Bertani (LB) broth (3–4 h) for the reactivation of BH4 strains before its testing.

2.4. Attenuated total reflection-Fourier transform infrared (ATR-FTIR) analysis

The ATR-FTIR spectroscopy was used to investigate the change in the membrane surface composition after the hydrophilic polymer coating and BH4 cell immobilization [35]. The PS and PVDF membrane samples (1-cm long each) were freeze-dried overnight and analyzed by a PerkinElmer FTIR/NIR spectrophotometer. The FTIR spectra were collected in the 4000–400 cm⁻¹ region with a resolution of 4 cm⁻¹ in the ATR mode.

2.5. Field emission-scanning electron microscope (FE-SEM) analysis of membrane surfaces

FE-SEM analyses of naked (as prepared or obtained without coating), vacant, and QQ membrane specimens (1-cm long) were conducted with slight modification [25]. More specifically, vacant and QQ membranes were first fixed with 2.5% (v/v) glutaraldehyde (Sigma Aldrich) in a 0.1-M phosphate buffer (pH of 7) for 2 h at 4 °C. The membranes were cleansed three times in a 0.1-M phosphate buffer (pH of 7) to eliminate any leftover glutaraldehyde. The membranes were dehydrated using ethanol solutions, with a stepwise increase in ethanol content (30%, 60%, 90%, and 100%). Subsequently, the membranes were freeze-dried overnight (Eyela Freeze Dryer FD-1000, Japan), sputtered with Pt, and finally examined on an SEM (SU-8220, Hitachi, Japan). Besides, energy-dispersive X-ray analysis was performed to determine the elemental compositions of the membrane surfaces.

2.6. Determination of QQ activities

The QQ activities of PS and PVDF membranes were evaluated in terms of the degradation rate of N-octanoyl-L-homoserine lactone (C8-HSL) over time. Several pieces of naked, vacant, and QQ membranes (each of 15-cm² surface area) were placed in 30 mL of 50-nM C8-HSL

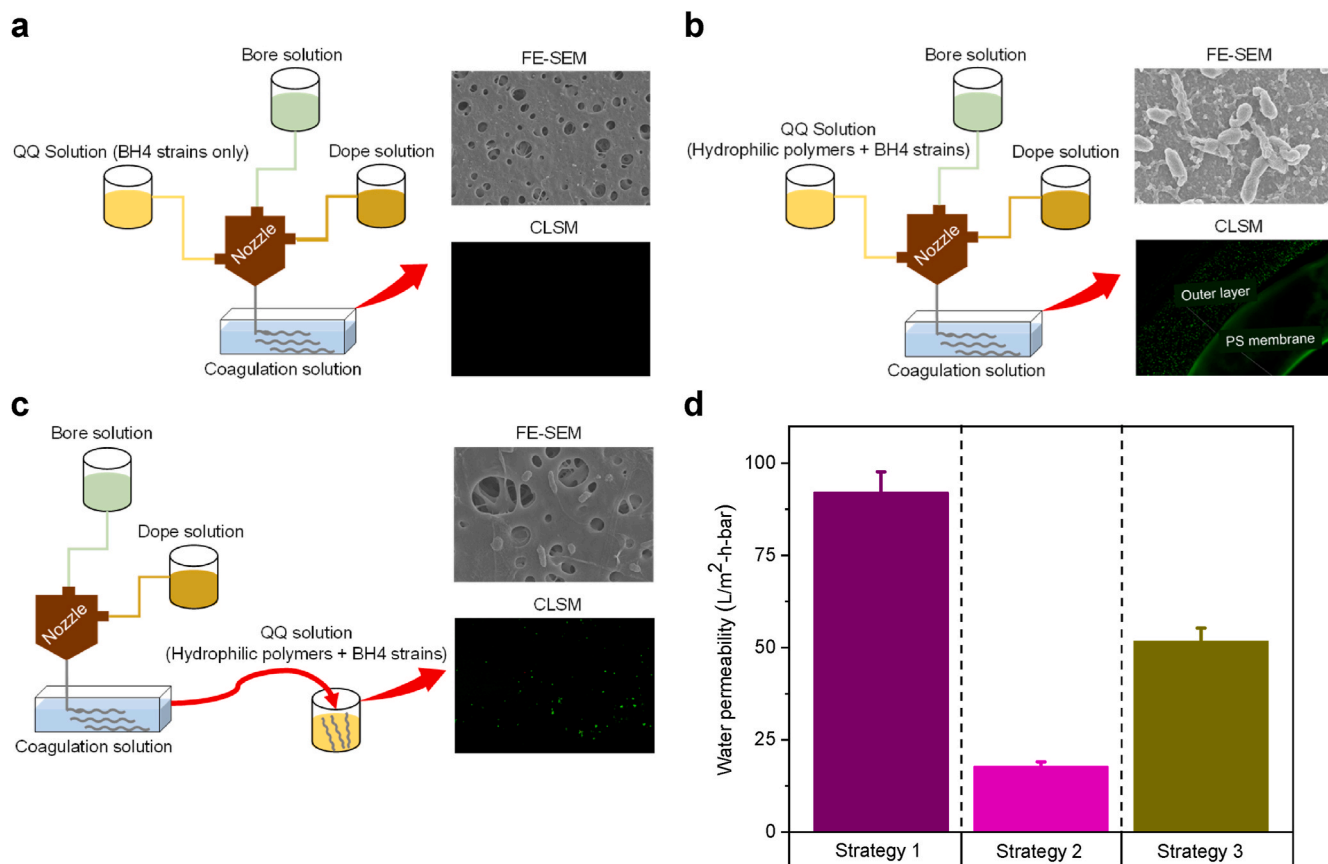


Fig. 1. Schematic representation of the strategies to immobilize BH4 strains on the membrane surface. (a) Strategy 1, where membrane formation was conducted with simultaneous BH4 immobilization. The QQ solution contains only BH4 strains. (b) Strategy 2 is similar to Strategy 1 except that the QQ solution contains BH4 strains and hydrophilic polymers. (c) Strategy 3, where the membrane was first formed and then immersed in the QQ solution containing BH4 strains and hydrophilic polymers. (d) Water permeabilities of the QQ membranes prepared by Strategy 1–3.

solution, and an aliquot (500 μ L) of each solution was collected at 0, 15, 30, 60, 90, 120, 150, 180, 240, and 300 min for bioassays while swirling the solution at 100 rpm in a shaker at 30 $^{\circ}$ C. The C8-HSL concentration of each sample was quantified through a bioassay using indicating agar plates [8,36]. A reporter strain (*Agrobacterium tumefaciens* A136) was grown in the LB broth containing the two antibiotics (tetracycline and spectinomycin). The indicating agar plate was prepared by putting the functional layer-containing LB agar (9 mL), two antibiotics, 1-mL A136, and X-gal (5-bromo-4-chloro-3-indolyl β -D-galactopyranoside) on top of a supporting layer (10 mL of LB agar). Holes (10 mm) were made in each agar plate using sterilized Pasteur pipettes. Each hole was then loaded with 20 μ L of each sample (collected from the aliquot). Finally, the agar plate was incubated for 24 h at 30 $^{\circ}$ C. The unknown C8-HSL concentrations were determined based on the diameter of the developed blue-colored circles after calibration using known C8-HSL concentrations [8]. Also, the indicating agar plates were directly used to visualize the QQ activity of membranes by placing and cultivating the membrane (3-cm each) on top of the plate at 30 $^{\circ}$ C for 24 h.

2.7. Confocal laser scanning microscope (CLSM) analysis for live QQ bacteria on the membrane surface

The live (green) or dead (red) cells of BH4 strains on the membrane surfaces were visualized using a CLSM (LSM 700, Carl Zeiss, Germany). Naked, vacant, and QQ membranes were stained (each 5-mm long) with a Live/Dead BacLight Bacterial Viability kit (L7012 Live/Dead BacLight, Molecular Probes, USA) for 15 min in the dark and observed under a microscope after gently washing the samples with deionized water three times to remove the residue. The CLSM images were analyzed using the

ImageJ software (<http://rsbweb.nih.gov>) and the Comstat2 program (<http://www.comstat.dk>) was used to quantitatively calculate the bio-volume (the volume of all voxels that contain biomaterials divided by the area) of live BH4 cells after staining along the depth of each image [37,38].

2.8. Examination of biofilms and extracellular biopolymers

The production of biofilms and extracellular polymeric substances was examined by co-culturing each membrane sample (0.628 cm^2) with a model QS bacterium PAO1 for 24 h at 30 $^{\circ}$ C [39]. For visualization, two reagents were applied as follows: (i) SYTO9 from the Live/Dead BacLight Bacterial Viability kit (L7012 Live/Dead BacLight, Molecular Probes, USA) to target all live bacterial cells and (ii) concanavalin A (Con A) conjugated with tetramethylrhodamine (Molecular Probes, USA) to target the polysaccharides with D-glucose or D-mannose. The membrane sample retrieved after culture was placed and stained in a conical tube (1.7 mL) at room temperature for 15 min. After staining, all samples were washed three times using clean water to remove the residue. The stained membrane was mounted on a CLSM (LSM 700, Carl Zeiss, Germany) for the observation of live cells and extracellular polymeric substances. The fluorescence of SYTO9 was detected at 488-nm excitation and 505–530-nm emission, whereas that of polysaccharides stained with ConA–tetramethylrhodamine was detected at 543-nm excitation and 550–600-nm emission [40,41]. The sample images were observed using a 200 $\times$ objective lens and the ZEISS ZEN Imaging Software. The biovolumes of the bacterial cells grown (green) and polysaccharides excreted (red) were calculated as described above.

2.9. Membrane characterization and biofouling tests

The pore size and porosity of membranes were determined using a mercury intrusion porosimeter (AutoPore V 9600, Micromeritics, USA). The dynamic contact angle of membranes was measured with time using a video microscope (SV-35, Somatech, Korea) to check the surface hydrophilicity.

Pure water permeabilities of the membranes used were tested using deionized water at a constant flux of 15 L/m²·h and a water temperature of 25 °C. Before the permeability test, the membranes were soaked in water for 30 min and then pressurized using N₂ gas at 70 kPa for another 5 min to check for leakage in the membrane module. The water permeability (in L/m²·h·bar) is expressed as the water flow rate per unit area of membrane per unit pressure [42].

Membrane biofouling tests were conducted in bioreactors employing three membrane filters, such as naked, vacant, and QQ membranes. Fig. 2 illustrates three MBRs with a working volume of 2 L each, which were tested in parallel under the conditions given in Table 1. The volume of each reactor was maintained at 2 L using an electric level sensor, whereas the reactor temperature was maintained at 25 °C by circulating water from a temperature-controlled water bath through the reactor jacket. The hydraulic retention time of the three bioreactors was set to 7.86 h at a flux of 15 L/m²·h and then reduced with the flux increase accordingly. However, the solids retention time was maintained at 100 d [9]. The transmembrane pressure (TMP) of each reactor was recorded continuously with a data logging system.

For the PS membranes, the three reactors were operated at a constant flux (15 L/m²-h) with a cycle of 9-min filtration and 1-min relaxation, along with continuous air sparging at 1 L/min (corresponding to 72 s⁻¹ in velocity gradient) underneath the membrane module [24]. Membrane filtration lasted until the TMP reached ~50 kPa. Two replicate experiments (Run 1 and 2) were performed, using virgin membranes. For PVDF membranes, membrane filtration started at a flux of 15 L/m²-h and lasted for ~8 d under the same conditions and procedures as described above (Phase 1). Then, the flux level was increased to 20 L/m²-h and operated for another 11 days (Phase 2). The membrane flux

Table 1

Summary of the operational conditions for the MBRs.

Run or Phase	Operation time (d)	Flux (L/m ² ·h)	Membrane type		
			Reactor 1	Reactor 2	Reactor 3
1	0–15	15	Naked, PS	Vacant, PS	QQ, PS
2	15–30	15	Naked, PS	Vacant, PS	QQ, PS
1	0–7	15	Naked, PVDF	Vacant, PVDF	QQ, PVDF
2	7–19	20	Naked, PVDF	Vacant, PVDF	QQ, PVDF
3	19–38	25	Naked, PVDF	Vacant, PVDF	QQ, PVDF

was further increased to 25 L/m²-h and operated until the TMP reached ~50 kPa.

3. Results and discussion

3.1. Strategies for the immobilization of live QQ bacterial strains on membrane filters

Several strategies to immobilize live QQ bacterial strains onto polymeric membrane filters were developed and are illustrated in Fig. 1. In the first attempt (Strategy 1), the possibility for the direct attachment of BH4 strains during the formation of hollow fiber membranes by phase inversion was explored (details are provided in the Methods section). Membrane pores were created in the coagulation solution by phase inversion. However, no BH4 cells were attached to the membrane surface (Fig. 1a). BH4 cells present in the QQ solution (aqueous phase) must be expelled from the hydrophobic solid (membrane) matrix formed by phase inversion. The result indicates that immobilizing the BH4 cells at the membrane surface is unachievable by Strategy 1.

In the second trial (Strategy 2), the QQ solution, consisting of hydrophilic polymers (alginate and PVA) and BH4 cells, was used. Here, dual-layered hollow fiber membranes were produced. An outer layer entrapping the BH4 cells, which are observed on the FE-SEM image and

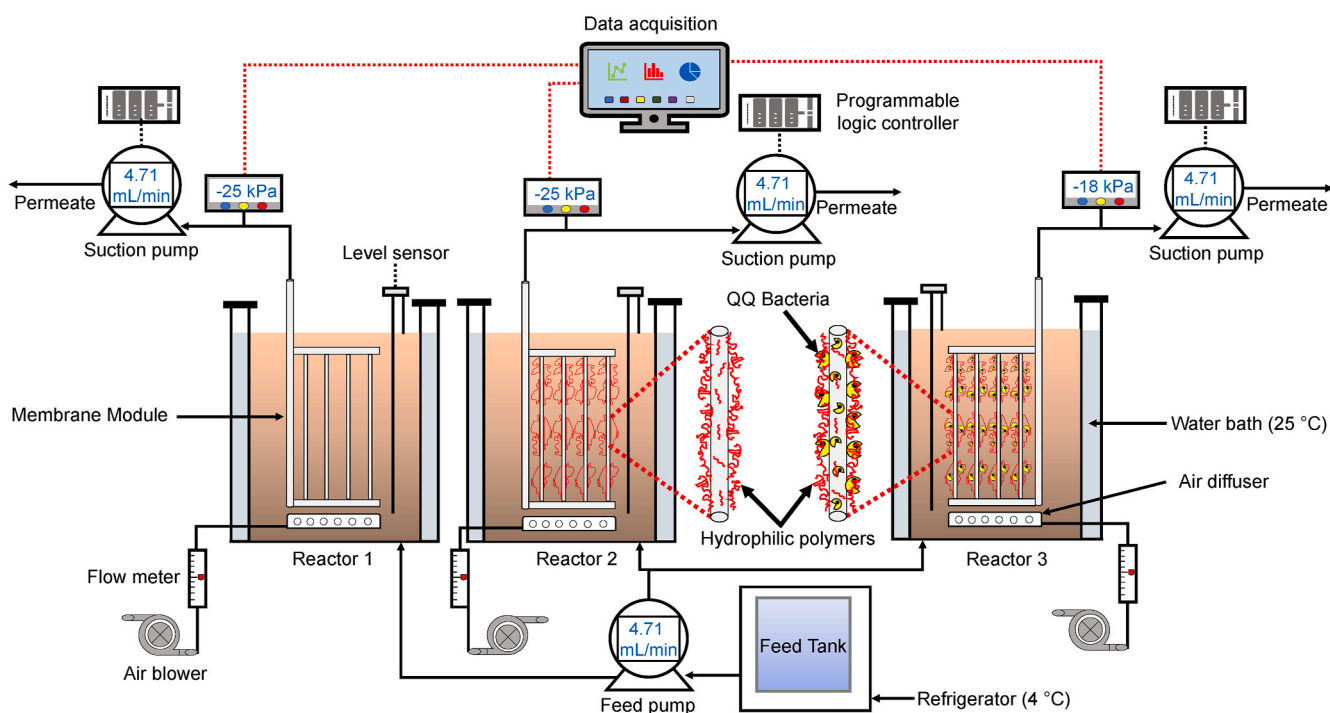


Fig. 2. Schematic representation of the MBRs used for membrane biofouling tests. The MBRs were operated with naked, vacant, and QQ (PS and PVDF) membranes. Experimental details are provided in [Table 1](#).

shown in green on the CLSM image inside the hydrophilic polymer matrix (Fig. 1b), was formed, surrounding the inner PS hollow fiber. With this strategy, however, the water permeability of the membrane significantly decreased to $<25 \text{ L/m}^2\text{-h-bar}$, which is extremely low for practical use. The double-layered membrane structure should offer more resistance to permeation.

In Strategy 3, BH4 cell immobilization was separately implemented by placing the membrane fiber in the QQ solution containing hydrophilic polymers and BH4 strains after membrane formation. BH4 bacteria were found attached to the membrane surface and remained alive as observed in the FE-SEM and CLSM images (Fig. 1c). Water permeability declined with cell immobilization (Fig. 1d), but its level ($\sim 50 \text{ L/m}^2\text{-h-bar}$) was acceptable for use ($>30 \text{ L/m}^2\text{-h-bar}$). Thus, QQ membranes with live BH4 cells attached are fabricated by Strategy 3, which can be extended for surface modification of commercially available membranes, and will be further discussed in later sections.

3.2. Spectroscopic and morphological characteristics of membrane filters before and after hydrophilic polymer coating and QQ cell immobilization

Fig. 3a and b show the ATR-FTIR spectra for lab-made PS and

commercial PVDF membrane filters before (designated as naked membranes) and after coating the hydrophilic polymers (a mixture of alginate and PVA) alone (designated as vacant membranes) and immobilizing the live QQ cells with the polymers (designated as QQ membranes) by Strategy 3. For the lab-made PS membranes, the characteristic peaks associated with aryl ether [43] and sulfonic groups [44] were identified at 1500 , 1320 , and 1150 cm^{-1} , respectively. The $\text{C}=\text{O}$ stretching peak [45] of the additive (polyvinylpyrrolidone) was identified near 1660 cm^{-1} . The key characteristic peaks of hydrophilic polymers (PVA), for the symmetrical stretching vibration of $-\text{OH}$ and asymmetric stretching vibration of $-\text{CH}_2$, respectively [46], were identified at 3320 and 2933 cm^{-1} . For commercial PVDF membranes, the characteristic peaks, such as $-\text{CH}_2$ wagging, $\text{C}-\text{C}$ stretching, and $\text{C}-\text{F}$ stretching, were detected at 1403 , 1185 , and 840 cm^{-1} , respectively [47,48]. After the hydrophilic polymer coating and cell immobilization, the $-\text{OH}$ and $-\text{CH}_2$ peaks were also detected as discussed above. Using the atomic percentage values of S and F elements of the PS and PVDF membrane surfaces (Fig. S1), the surface coverage by hydrophilic polymers was estimated to be 40–60% and 50–65%, respectively. The results confirm that hydrophilic polymers bind well to the naked PS and PVDF membranes when they form vacant and QQ membranes by

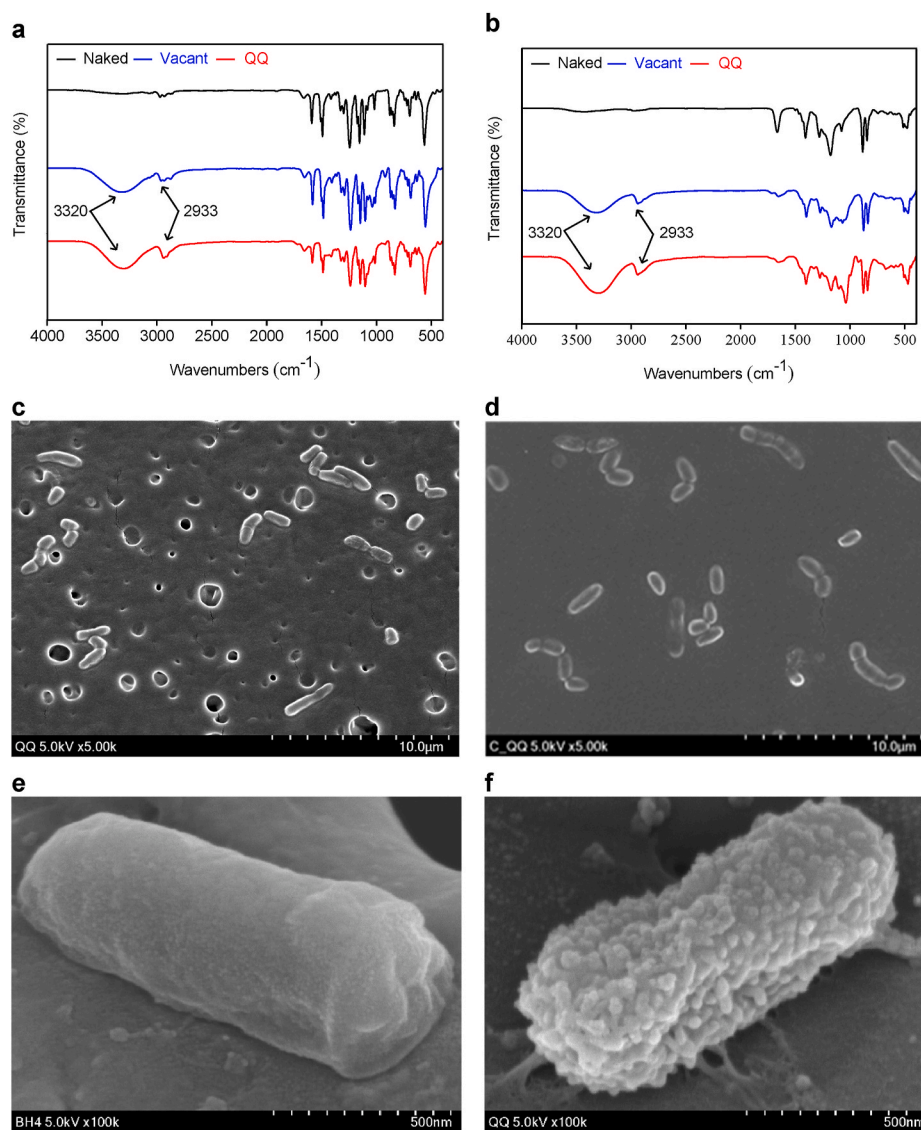


Fig. 3. Spectroscopic and morphological characteristics of membrane filters. ATR-FTIR spectra of various (a) PS and (b) PVDF membranes. FE-SEM images of (c) PS and (d) PVDF QQ membranes. FE-SEM images of a BH4 lying on a membrane (e) without and (f) with hydrophilic polymers.

Strategy 3.

The FE-SEM photos show the untouched surfaces of vacant and QQ membranes (Fig. 3c and d, and Fig. S2). Some morphological changes exist after the coating of the PS membrane using hydrophilic polymers (vacant PS membrane), because the polymers were deposited on the membrane surface, as confirmed in the ATR-FTIR analysis. For the QQ PS membrane, the rod-shaped BH4 bacteria immobilized together with hydrophilic polymers were observed on the membrane surface. Similarly, morphological changes that occurred on the PVDF membrane were observed with the coating of hydrophilic polymers. Its pore openings, smaller than those of the PS membrane, were covered by the polymers. The BH4 strains were also observed on the QQ membrane. In addition, it was observed that hydrophilic polymers were deposited (which formed bumpy cell surfaces), which simultaneously anchored the BH4 strain to the membrane surface, as opposed to the smooth surface of a pure BH4 strain (Fig. 3e and f). The coated polymer layer

thickness on top of the vacant and QQ membranes was estimated to be in the range of 1.5–3.5 μm (Table S1). There were no significant changes in the surface hydrophilicity, pore size, and porosity of membranes before and after coating, although the coated membranes became wet a lot more quickly than the naked ones, probably due to hydrophilic polymers (Fig. S3). The result indicates that hydrophilic polymers play a key role in the immobilization of BH4 strains to the membrane.

3.3. Biochemical activities of live QQ cell-immobilized membranes

Fig. 4a and b visualize the live BH4 cells immobilized on the membrane surfaces as indicated in green on the QQ PS and PVDF membranes. The naked and vacant membrane surfaces, devoid of live cells, are all shown in black (Fig. S4). The quantity of live BH4 strains on the QQ PS and PVDF membranes, defined as the biovolume (calculation details are provided in Section 2.7), were estimated to be 0.0044 and 0.0027 $\mu\text{m}^3/$

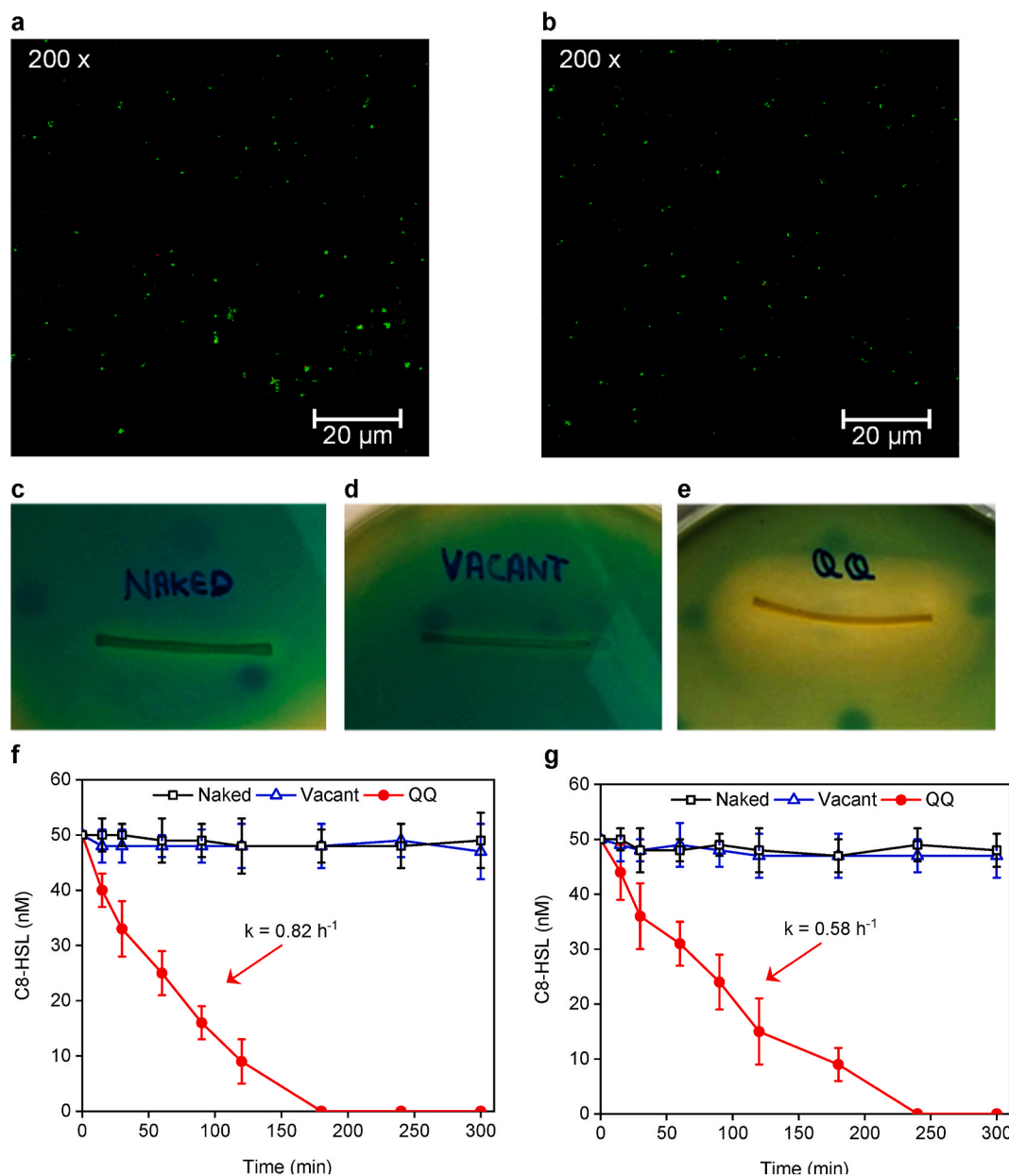


Fig. 4. Biochemical characteristics of QQ membranes. CLSM images of (a) PS and (b) PVDF QQ membranes after staining with a Live/Dead BacLight Bacterial Viability kit. Photos of (c) naked, (d) vacant, and (e) QQ PS membrane fibers placed on QQ bioassay agar plates. C8-HSL degradation profiles of (f) PS and (g) PVDF QQ membranes. Error bars indicate one standard deviation (n = 2).

μm^2 , respectively (Fig. S5). The result demonstrates the successful immobilization of live BH4 strains on the two membranes, using hydrophilic polymers.

The QQ activity of live BH4 cell-immobilized membranes was then determined by the bioassay using the LB agar plate containing the signal molecule, C8-HSL, and the reporter strain, *Agrobacterium tumefaciens* A136 (Fig. 4c–e). The naked and vacant hollow fiber membranes placed on the LB agar plates turned greenish after a 24-h incubation, indicating that no signal molecule degradation occurred by the membranes. In contrast, the green-color disappeared near the QQ membrane fiber,

revealing that signal molecules around the membrane were well degraded by the live BH4 immobilized on the membrane surface. Besides, the same behaviors were observed with the PVDF membranes (Fig. S6). A previous study reported that the enzymatic inactivation of signal molecules (i.e., C8-HSL), which should be otherwise detected in the membrane biocake, was achieved when porcine kidney acylase I was added to an MBR [36].

The degradation rate of the dominant C8-HSL was further evaluated in aqueous media (Fig. 4f and g). No C8-HSL degradation by the naked and vacant membranes occurred during the 300-min tests irrespective of

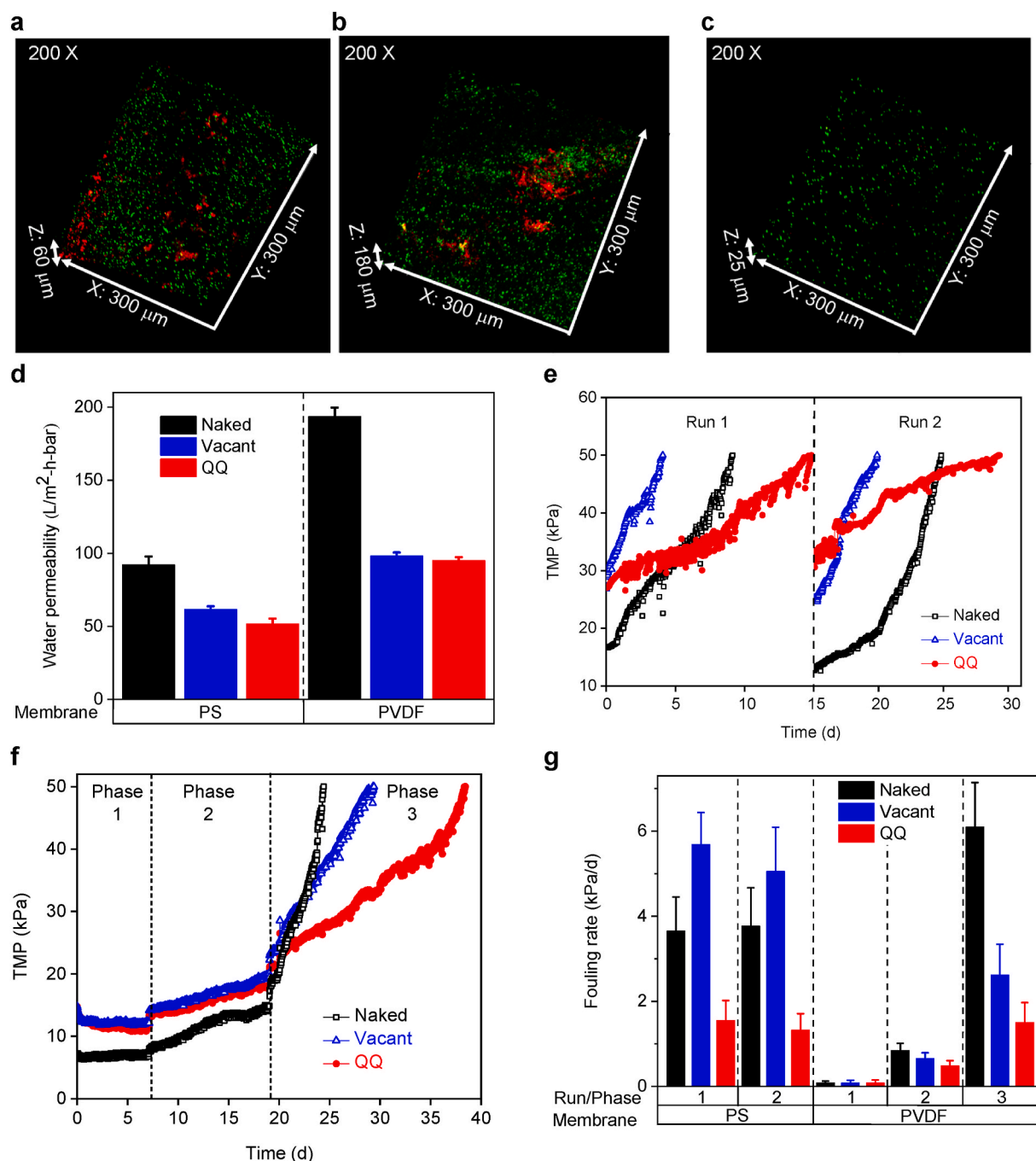


Fig. 5. Membrane biofilms and biofouling tests. CLSM images of live bacterial cells (green) and polysaccharides (red) on (a) naked, (b) vacant, and (c) QQ PS membranes after co-culture with a model bacterium (PAO1) for 24 h at 30 °C. (d) Pure water permeabilities of various PS and PVDF membranes. Error bars indicate one standard deviation ($n = 2$). TMP build-up profiles of various (e) PS and (f) PVDF membranes with time under different operation conditions. Operational details are provided in Table 1. (g) Respective average fouling rate for each TMP profile of Fig. 5e and f (error bars represent a 95% confidence level). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

membrane types. However, its degradation was significant and completed within 180 and 240 min by the QQ PS and QQ PVDF membranes, respectively. The degradation first-order rate constants of the signal molecule by the former and latter were estimated to be 0.82 and 0.58 h⁻¹, respectively. The magnitude of the rate constants seems to be closely related to the biovolume (quantity) of live BH4 cells present on the membrane surface. Thus, the difference in the degradation rate stems from the amount of BH4 cells immobilized on the QQ membrane surfaces. The result is in agreement with a previous finding on QQ media, whose activity was proportional to the QQ bacteria dose [9].

3.4. Membrane biofilm formation behaviors

The biofilm formation behaviors for the six membranes were tested using a representative biofilm-forming bacterium, *P. aeruginosa* PAO1 (Fig. 5a–c and Fig. S7). Relatively substantive amounts of PAO1 biofilms (in green) and polysaccharide biopolymers (in red) were observed on the naked and vacant membranes, where no QQ bacteria were immobilized. Green spots were also observed on the QQ membranes due to the BH4 cells immobilized on them. The biovolumes of the green spots before and after the biofilm formation tests slightly increased from 0.0044 to 0.0067 μm³/μm² and 0.0027–0.0028 μm³/μm² for the QQ PS and PVDF membranes, respectively, which are considerably lower than those of the naked and vacant membranes (Fig. S8). Besides, the biovolumes of the biopolymer polysaccharides were also significantly smaller with the QQ membranes than those of the naked and vacant membranes. Red colors on the QQ membrane were almost invisible (Fig. 5c and Fig. S7c). The result confirms that the BH4 strains immobilized on the QQ membranes interfere with the growth of PAO1 biofilms as those in QQ media did in previous studies [30]. The overall trends in both membranes (PS and PVDF) were almost the same except for the relatively sparse distributions of green and red spots on the PVDF membrane, which can be attributed to the membrane material characteristics. PVDF (34.4 mJ/m²) has lower surface free energies than PS (46.6 mJ/cm²) [49], so the adhesion onto the former would be relatively less. Consequently, live BH4 cells immobilized on the QQ membranes degrade the signal molecules produced by PAO1 [50], inhibiting biofilm formation and subsequent biopolymer secretion. The result exhibits the potential of QQ membranes for anti-biofouling efficacy.

3.5. Membrane anti-biofouling performance in bioreactors

Fig. 5d shows the variations of water permeability of PS and PVDF membranes when the hydrophilic polymers were coated and BH4 cells were immobilized on them. The water permeability values of the vacant and QQ PS membranes decreased by 34.0% and 43.6% of that of the naked one, respectively. Similarly, the water permeability reductions of the vacant and QQ PVDF membranes were more significant than those of the vacant and QQ PS membranes and were estimated to be ~46.3% and 47.0% compared to that of the naked PVDF one, respectively. Nevertheless, the water permeability values (>50 L/m²-h-bar) of all prepared membranes were acceptable for actual filtration applications [51].

Fig. 5e shows the TMP profiles of six membrane filters employed for MBRs operated at constant flux levels between 15 and 25 L/m²-h. The naked PS membrane had a relatively low TMP value of ~16 kPa at a membrane flux of 15 L/m²-h at the beginning of membrane operation, whereas that of the vacant and QQ membranes was 10 kPa higher (~26 kPa). The disparity is because the coating layers (hydrophilic polymers and/or QQ bacteria) resist water permeation as discussed above. With continuous membrane filtration (Run 1), the naked and vacant membranes experienced a rapid TMP increase, with their TMP levels reaching 50 kPa within 4.5 and 9.5 days, respectively. In contrast, the QQ membrane was the slowest in TMP increment (~14.6 days) than the other two until it reached 50 kPa. Another replicate test (Run 2 of Fig. 5e) showed similar trends in TMP increment. The commercial PVDF membranes were also tested in the same manner. However, the

membrane flux increased stepwisely from 15 to 25 L/m²-h over time (Fig. 5f). During the first seven days of membrane operation (Phase 1), the three PVDF-based membranes (naked, vacant, and QQ membranes) showed no significant TMP increment over time, although the two coated membranes (vacant and QQ membranes) had higher TMP levels because of the additional coating layers. The relatively low TMP levels of the commercial PVDF membranes compared to those of the lab-made PS membranes reflect higher water permeability of the former than the latter. When the flux level increased from 15 to 20 L/m²-h (Phase 2), the TMP level began to increase gradually. After the flux level was further stressed to 25 L/m²-h on the 20th day of the membrane operation, the TMP increment became significant, particularly for the naked and vacant membranes. However, the QQ membrane was slower in TMP increment than the other two. As summarized in Fig. 5g, the live BH4 strains on the membrane surface delayed the rate of membrane biofouling by 57%–67% and 30%–57% compared to the naked and vacant membranes, respectively. This result demonstrates that live BH4 cells immobilized on the membrane mitigate biofouling as effectively as those in QQ media alleviated membrane biofouling in previous studies [19,52].

In summary, live anti-biofouling membrane filters are successfully fabricated through their immersion into the solution containing BH4 strains and hydrophilic polymers (Strategy 3). The live BH4 cells immobilized on the membrane can degrade the C8-HSL, which inhibits biofilm formation and consequently, mitigates membrane biofouling. The strategy to immobilize live bacterial cells on the membrane filter is potentially applicable in several areas associated with quorum sensing and quenching in the industry and our daily lives.

4. Conclusions

BH4 strain-immobilized membrane filters were fabricated by several strategies and characterized to investigate their spectroscopic, morphological, and biochemical properties. Furthermore, they were tested to determine their anti-biofouling efficacy in MBRs. We can thus conclude the followings:

- (1) Among the three key approaches devised for the fabrication of a QQ membrane, the immersion and incubation of a membrane filter into the QQ solution containing a mixture of BH4 strains and hydrophilic polymers (Strategy 3) enabled BH4 cells to immobilize on the lab-made PS and commercial PVDF membrane filters.
- (2) Live BH4 cell-immobilized membranes degraded the C8-HSL, showing their QQ activities at rate constants of 0.82 and 0.58 h⁻¹ for the PS and PVDF QQ membranes, with BH4 biovolumes of 0.0044 and 0.0027 μm³/μm², respectively.
- (3) The biofilm-forming bacterium PAO1 grew and produced biopolymers on the naked and vacant membranes. However, its growth and subsequent biopolymer secretions on the QQ membranes were markedly inhibited.
- (4) Accordingly, the membrane biofouling rates were significantly mitigated for both the lab-made PS and commercial PVDF QQ membranes, which apply to the continuous operation of MBRs, treating synthetic wastewater.
- (5) Some challenges remain to be resolved in future studies, such as minimizing the water permeability loss with the immobilization of live QQ bacterial strains and maintaining the activity and stability of QQ membranes during long-term use.

Author statement

Syed Salman Ali Shah: Methodology, Investigation, Data Curation, and Writing - Original Draft. Kibeak Lee: Methodology and Investigation. Hyeona Park: Methodology and Investigation. Kwang-Ho Choo: Conceptualization, Writing - Review & Editing, Supervision, and

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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